HEADLINE

GRAPHICS CARDS

Enough is not enough when it comes to 3D graphics cards. Before you take the leap and buy the latest card with truck-loads of memory and breakneck clock speeds—besides the promise of a shattering on-screen gaming experience—ask yourself how much power do you really need and, more importantly, do you have the system to support it? The newer cards based on the GeForce 3 and GeForce 4 chipsets need a high-horsepower system to belt out the ultimate gaming experience. If you have anything less than a 1 GHz-based system with at least 256 MB ram and a 17-inch monitor, you are not doing justice to the power of these cards.

The number of games that support the high-end features offered by the top cards of today is the lower side too. This brings us to the value segment cards based on the various GeForce2 MX chipsets (MX-200, MX400, and MX). These cards are priced a lot lower and yet, offer a reasonably good gaming experience at normal to high resolutions.

Most manufacturers provide additional ports like Dual-Display and TV-Out S-Video in their cards. Useful software and games are also bundled along with these cards—this makes them a great buy. In this test, we compare 21 such value-added cards that hit the sweet spot without burning a hole in your pocket.

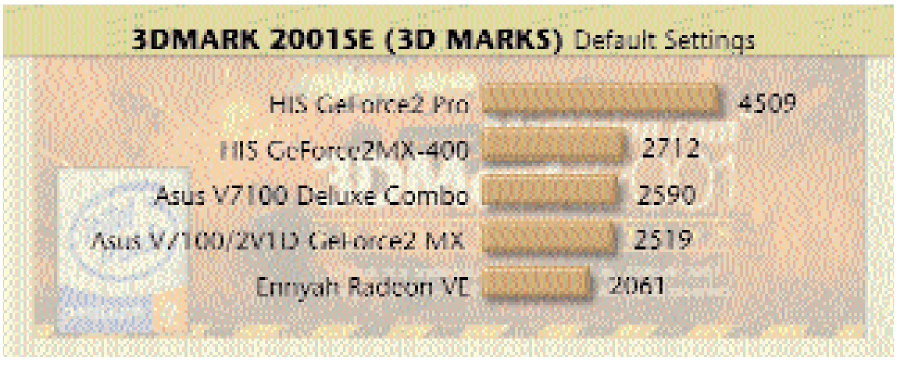
Overview of the cards

A plethora of value-based cards found their way into our comparison test, ranging from the old M64-based cards to the formidable GeForce2MX chipsets. Most of the MX-based cards featured additional ports like TV-Out, S-Video in/out, and so on. The Matrox cards sported dual-display ports. We also received cards based on the different variants of the ATi Rage 128 chipset.

Synthetic tests

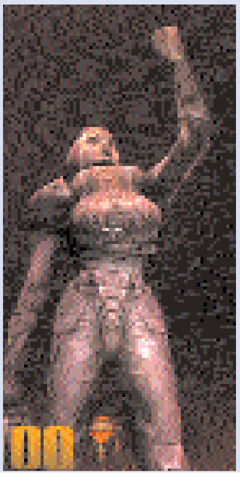
Our first round of tests included synthetic benchmarks which test the performance of 3D cards on individual parameters. In 3DMark 2001SE, the cards that failed the test altogether were the Matrox G450 and the Maxforce Riva TNT M64. The Matrox card was the 16-MB PCI based version and it failed this test because the test by itself requires xxx MB memory.

The cards that did well in this test were the ones based on ATi Radeon and GeForce2 MX chipsets. The Asus card to top the score chart was the V7100 Deluxe Combo—it scored 2590. Competition came from the Radeon-based Ennyah Radeon VE, which also performed pretty well to log a score of 2061. But the cards that really stole the show were the HIS GeForce2MX-400 and the HIS GeForce2



Pathe breaking features

Increasing frame rates alone is one thing and breathing visual reality into games is another. Leading chip manufacturers try hard to bring out features in their chips that make 3D games look more lifelike. But, till now, only a few have made any major impact. Here are some such features that helped the cause.



Anti-aliasing: This feature was introduced by 3dfx. Simply speaking anti-aliasing means reducing the 'jaggies' that you see on the edges of textures in a 3D game. Modern cards support up to 4X AA. Enabling this feature does reduce frame rates considerably but the trade off is the near-perfect picture quality.

Anisotropic filtering: This is an advanced texture-filtering technique that greatly enhances image quality in scenes where objects extend deep into the background from the foreground.

Caption

Environment mapped bump mapping: Introduced by Matrox, this is a shading technique that uses multiple textures and lighting effects to simulate wrinkled or bumped surfaces. Previously, textures in games never used to cast their own shadows or reflections, for example, ripples in a pond. But with bump mapping, the surface detail of a 3D texture is brought to life.

Transform and Lighting: Introduced by nVidia, this feature uses two separate engines on the GPU for providing scenes with an extremely high polygon count without sacrificing on frame-rate. Transform determines how complex objects can appear in a scene without sacrificing the frame-rate. Lighting techniques, on the other hand, change the appearance of objects based on light sources.

nFiniteFX: Introduced by nVidia, this feature gives game developers the freedom to program a virtually infinite number of custom-made special effects like underwater refractions, facial expressions, etc. This means that game characters and envi-